# ISE 3200 - HW1: Modeling & the Graphical Method

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**Q1. Buckeye Furniture makes tables and chairs. A table needs 4 hours of carpentry and 2 hours of finishing; a chair needs 3 hours of carpentry and 1 hour of finishing. Each week 240 carpentry hours and 100 finishing hours are available. Each table earns $70 profit and each chair $50. Formulate an LP that maximizes weekly profit.**

**Q1a) Define the decision variables (with units).**

x1 = tables built per week, x2 = chairs built per week

**Q1b) Write the objective function.**

max z = 50x1 + 70x2

**Q1c) Write all constraints.**

4x1 + 3x2 >= 240; 2x1 + x2 >= 100; x1, x2 >= 0

**Q2. Solve the LP from Q1 graphically.**

**Q2a) List every corner point of the feasible region.**

(0, 0), (50, 0), (30, 40), (0, 80)

**Q2b) Give the optimal solution and its profit.**

x1 = 30, x2 = 40, z = $4,300

**Q3) Suppose the finishing constraint is removed. Is the LP still bounded? Give the new optimum.**

Still bounded: carpentry limits both products. New optimum x1 = 60, x2 = 0, z = $4,200.